

A box contains 6,400 balls. The balls increase by 5% and then decrease by 93.75%. How many balls remain?
 एक डिब्बे में 6,400 गेंदें हैं। गेंदों की संख्या में 5% की वृद्धि होती है और फिर 93.75% की कमी हो जाती है। अब कितनी गेंदें शेष रहेंगी?

(A) 6,300

(B) 400

(C) 480

~~(D) 420~~

(E) None of these

$$1) + 5\% \Rightarrow 1 + \frac{1}{20} \Rightarrow \frac{21}{20}$$

$$2) 93.75\% \Rightarrow 1 - 6.25\% \Rightarrow 1 - \frac{1}{16} \Rightarrow \frac{15}{16}$$

$$\# \left\{ 6400 \times \frac{21}{20} \right\} \times \frac{1}{16} \Rightarrow \underline{420}$$

$$- 93.75\% \Rightarrow 1 - \frac{15}{16} \Rightarrow \frac{1}{16}$$

The population of a town is 9,380. It decreases by 42.86% and then increases by 37.50%. Find the final population.

एक कस्बे की जनसंख्या 9,380 है। यह पहले 42.86% घटती है और फिर 37.50% बढ़ती है। अंतिम जनसंख्या ज्ञात कीजिए।

(A) 5,360

(B) 7,030

~~(C) 7,370~~

(D) 8,250

(E) None of these

$$1) 42.85\% = \frac{3}{7}$$

$$- 42.85\% \Rightarrow 1 - \frac{3}{7} = \frac{4}{7}$$

$$\# \left\{ 9380 \times \frac{4}{7} \right\} \times \frac{11}{8} \Rightarrow \underline{7370}$$

$$2) + 37.5\% \Rightarrow 1 + \frac{3}{8} \Rightarrow \frac{11}{8}$$

A person's salary is increased by 87.50% and then reduced by 28%. The final salary is Rs. 12,960. What was the original salary?

एक व्यक्ति के वेतन में पहले 87.50% की वृद्धि की जाती है और फिर 28% की कमी की जाती है। अंतिम वेतन 12,960 रुपये है। मूल वेतन क्या था?

(A) Rs. 8,640

~~(B) Rs. 9,600~~

(C) Rs. 10,000

(D) Rs. 11,520

(E) None of these

$$1) 87.5\% = \frac{7}{8}$$

$$+ 87.5\% = 1 + \frac{7}{8} \Rightarrow \frac{15}{8}$$

$$\# \left\{ \text{Original} \times \frac{15}{8} \right\} \times \frac{18}{25} \Rightarrow 12960$$

$$2) 28\% = (4\%) \times 7 = \frac{1}{25} \times 7 = \frac{7}{25}$$

$$\text{Original} = 12960 \times \frac{25}{18} \times \frac{8}{15}$$

$$- 28\% = 1 - \frac{7}{25} \Rightarrow \frac{18}{25}$$

$$\text{Original} = \underline{9600}$$

The salary of a person is decreased by 22.22% and then increased by 10% to become Rs. 16,170. Find the original salary.

एक व्यक्ति के वेतन में पहले 22.22% की कमी की जाती है और फिर उसमें 10% की वृद्धि करके उसे 16,170 रुपये कर दिया जाता है। मूल वेतन ज्ञात कीजिए।

(A) 17,900

(B) 18,900

(C) 19,800

(D) 18,000

(E) None of these

$$1) 22.22\% = (11.11\%) \times 2 \\ = \frac{1}{9} \times 2 = \frac{2}{9}$$

$$\# \left\{ \text{original} \times \frac{7}{9} \right\} \times \frac{11}{10} = 16170$$

$$- 22.22\% = 1 - \frac{2}{9} \Rightarrow \frac{7}{9}$$

$$\text{original} = 16170 \times \frac{10}{11} \times \frac{9}{7}$$

$$2) 10\% = \frac{1}{10}$$

$$\text{original} = \underline{18900}$$

$$+ 10\% = 1 + \frac{1}{10} = \frac{11}{10}$$

After being decreased by 55.56% and then increased by 18.18%, the amount becomes Rs. 13,000. Find the original amount.

55.56% की कमी और फिर 18.18% की वृद्धि के बाद, राशि 13,000 रुपये हो जाती है। मूल राशि ज्ञात कीजिए।

(A) 23,100

(B) 22,250

(C) 24,000

(D) 24,750

(E) None of these

$$1) 55.56\% = (11.11\%) \times 5 \\ = \frac{1}{9} \times 5 = \frac{5}{9}$$

$$\# \left\{ \text{original} \times \frac{4}{9} \right\} \times \frac{13}{11} = 13000$$

$$- 55.56\% \Rightarrow 1 - \frac{5}{9} = \frac{4}{9} \checkmark$$

$$\text{original} = \frac{13000}{1000} \times \frac{11}{13} \times \frac{9}{4}$$

$$2) 18.18\% = (9.09\%) \times 2 \\ = \frac{1}{11} \times 2 = \frac{2}{11}$$

$$\text{original} = \underline{24750}$$

$$+ 18.18\% = 1 + \frac{2}{11} \Rightarrow \frac{13}{11} \checkmark$$

A number is 23,125. It is increased successively by 14.29%, 16.67%, and 20%. Find the final number.

एक संख्या 23,125 है। इसे क्रमशः 14.29%, 16.67% और 20% से बढ़ाया जाता है। अंतिम संख्या ज्ञात कीजिए।

(A) 34,910

(B) 37,000

(C) 35,800

(D) 38,000

(E) 33,000

$$1) 14.29\% = \frac{1}{7}$$

$$+ 14.29\% = 1 + \frac{1}{7} \Rightarrow \frac{8}{7}$$

$$\# \left[\left\{ 23125 \times \frac{8}{7} \right\} \times \frac{7}{6} \right] \times \frac{6}{5}$$

$$2) 16.67\% = \frac{1}{6}$$

$$+ 16.67\% = 1 + \frac{1}{6} \Rightarrow \frac{7}{6}$$

$$\Rightarrow \frac{23125 \times 8}{7} \times \frac{7}{6} \times \frac{6}{5}$$

$$\Rightarrow 4625 \times 8$$

$$3) 20\% = \frac{1}{5}$$

$$+ 20\% = 1 + \frac{1}{5} \Rightarrow \frac{6}{5}$$

$$\Rightarrow \underline{37000}$$

A salary is Rs. 17,920. It is increased successively by 12.50%, 11.11%, and 25%. Find the final salary.

एक व्यक्ति का वेतन 17,920 रुपये है। इसमें क्रमशः 12.50%, 11.11% और 25% की वृद्धि की जाती है। अंतिम वेतन ज्ञात कीजिए।

(A) 28,000

(B) 26,880

(C) 27,600

(D) 22,400

(E) 29,120

$$1) 12.50\% = \frac{1}{8}$$

$$+ 12.5\% = 1 + \frac{1}{8} \Rightarrow \frac{9}{8}$$

$$\# \left[\left\{ 17920 \times \frac{9}{8} \right\} \times \frac{10}{9} \right] \times \frac{5}{4}$$

$$2) 11.11\% = \frac{1}{9}$$

$$+ 11.11\% = 1 + \frac{1}{9} \Rightarrow \frac{10}{9}$$

$$\Rightarrow 17920 \times \frac{9}{8} \times \frac{10}{9} \times \frac{5}{4}$$

$$\Rightarrow \underline{28000}$$

$$3) 25\% = \frac{1}{4}$$

$$+ 25\% = 1 + \frac{1}{4} \Rightarrow \frac{5}{4}$$

A quantity is 37,800. It is decreased successively by 33.33%, 8.33%, and 9.09%. Find the final quantity.
 एक राशि 37,800 है। इसे क्रमशः 33.33%, 8.33% और 9.09% से घटाया जाता है। अंतिम राशि ज्ञात कीजिए।

(A) 25,200

(B) 23,100

(C) 21,000

(D) 20,000

(E) 19,800

$$1) 33.33\% = \frac{1}{3}$$

$$- 33.33\% = 1 - \frac{1}{3} \Rightarrow \frac{2}{3}$$

$$\# \left[\left\{ 37800 \times \frac{2}{3} \right\} \times \frac{11}{12} \right] \times \frac{10}{11}$$

$$2) 8.33\% = \frac{1}{12}$$

$$- 8.33\% = 1 - \frac{1}{12} \Rightarrow \frac{11}{12}$$

$$\Rightarrow 37800 \times \frac{2}{3} \times \frac{11}{12} \times \frac{10}{11}$$

$$\Rightarrow \underline{21000}$$

$$3) 9.09\% = \frac{1}{11}$$

$$- 9.09\% = 1 - \frac{1}{11} \Rightarrow \frac{10}{11}$$

A factory output is 46,800 units. It is decreased successively by 37.50%, 7.69%, and 5.56%. Find the final output.

एक कारखाने का उत्पादन 46,800 इकाइयाँ है। इसे क्रमशः 37.50%, 7.69% और 5.56% घटाया जाता है। अंतिम उत्पादन ज्ञात कीजिए।

(A) 24,000

(B) 26,000

(C) 27,200

(D) 25,500

(E) 25,000

$$1) 37.5\% = \frac{3}{8}$$

$$- 37.5\% = 1 - \frac{3}{8} \Rightarrow \frac{5}{8}$$

$$\# \left[\left\{ 46800 \times \frac{5}{8} \right\} \times \frac{12}{13} \right] \times \frac{17}{18}$$

$$2) 7.69\% = \frac{1}{13}$$

$$- 7.69\% = 1 - \frac{1}{13} \Rightarrow \frac{12}{13}$$

$$\Rightarrow 46800 \times \frac{5}{8} \times \frac{12}{13} \times \frac{17}{18}$$

$$\Rightarrow \underline{25500}$$

$$3) 5.56\% = \frac{1}{18}$$

$$- 5.56\% = 1 - \frac{1}{18} \Rightarrow \frac{17}{18}$$

After being decreased successively by 45.45%, 21.43%, and 5%, a value becomes 28,500. Find the original value.

किसी मान को क्रमशः 45.45%, 21.43% और 5% घटाने पर उसका मान 28,500 हो जाता है। मूल मान ज्ञात कीजिए।

(A) 66,500

(B) 70,000

(C) 72,000

(D) 68,250

(E) 75,000

$$1) 45.45\% = (9.09\%) \times 5 \\ = \frac{1}{11} \times 5 = \frac{5}{11}$$

$$\# (\text{original}) \times \frac{6}{11} \times \frac{11}{14} \times \frac{19}{20} = 28500$$

$$- 45.45\% = 1 - \frac{5}{11} \Rightarrow \frac{6}{11} \checkmark$$

$$\Rightarrow \text{original} = 28500 \times \frac{20}{19} \times \frac{14}{11} \times \frac{11}{6}$$

$$2) 21.43\% = (7.14\%) \times 3 \\ = \frac{1}{14} \times 3 \Rightarrow \frac{3}{14}$$

$$\Rightarrow \text{original} \Rightarrow \underline{70,000}$$

$$- 21.43\% = 1 - \frac{3}{14} \Rightarrow \frac{11}{14} \checkmark$$

$$3) - 5\% \Rightarrow 1 - \frac{1}{20} \Rightarrow \frac{19}{20} \checkmark$$

A number of coins is 17,820. It is increased by 88.89%, then decreased by 36.36%, then increased by 8.33%. Find the final number.

सिक्कों की संख्या 17,820 है। इसमें 88.89% की वृद्धि की जाती है, फिर 36.36% की कमी की जाती है, और फिर 8.33% की वृद्धि की जाती है। अंतिम संख्या ज्ञात कीजिए।

(A) 23,205

(B) 21,420

(C) 33,660

(D) 19,800

(E) None of these

$$1) + 88.88\% = 1 + \frac{8}{9} \\ \Rightarrow \frac{17}{9}$$

$$\# 17820 \times \frac{17}{9} \times \frac{7}{11} \times \frac{13}{12}$$

$$2) - 36.36\% = 1 - \frac{4}{11} \\ \Rightarrow \frac{7}{11}$$

$$\Rightarrow 119 \times 13$$

$$\Rightarrow \underline{23205}$$

$$3) + 8.33\% = 1 + \frac{1}{12} \\ \Rightarrow \frac{13}{12}$$

A number is 35,700. It is increased by 15.38%, then decreased by 21.42%, then decreased by 13.33%. Find the final number.

एक संख्या 35,700 है। इसे पहले 15.38% बढ़ाया जाता है, फिर 21.42% घटाया जाता है, और फिर 13.33% घटाया जाता है। अंतिम संख्या ज्ञात कीजिए।

(A) 31,500

(B) 29,925

(C) 24,750

(D) 27,500

(E) None of these

$$1) 15.38\% = (7.69\%) \times 2 \\ = \frac{1}{13} \times 2 \Rightarrow \frac{2}{13}$$

$$\# \left[\left\{ 35700 \times \frac{15}{13} \right\} \times \frac{11}{14} \right] \times \frac{13}{15}$$

$$+ 15.38\% = 1 + \frac{2}{13} \Rightarrow \frac{15}{13} \checkmark$$

$$\Rightarrow 35700 \times \frac{15}{13} \times \frac{11}{14} \times \frac{13}{15}$$

$$2) -21.42\% = 1 - \frac{3}{14} \Rightarrow \frac{11}{14} \checkmark$$

$$\Rightarrow 35700 \times \frac{11}{14}$$

$$3) -13.33\% = 1 - \frac{2}{15} \Rightarrow \frac{13}{15} \checkmark \Rightarrow \underline{28050}$$

An amount is increased by 86.67%, then decreased by 6.25%, then decreased by 26.67%. If the final amount is Rs. 23,177, find the initial amount.

एक राशि में 86.67% की वृद्धि की जाती है, फिर 6.25% की कमी की जाती है, फिर 26.67% की कमी की जाती है। यदि अंतिम राशि 23,177 रुपये है, तो प्रारंभिक राशि ज्ञात कीजिए।

(A) Rs. 18,000

(B) Rs. 18,060

(C) Rs. 18,250

(D) Rs. 18,450

(E) None of these

$$1) + 86.67\% = 1 + \frac{13}{15} \\ \Rightarrow \frac{28}{15} \checkmark$$

$$\# \text{ original} \times \frac{28}{15} \times \frac{15}{16} \times \frac{11}{15} = 23177$$

$$2) -6.25\% = 1 - \frac{1}{16} \Rightarrow \frac{15}{16} \checkmark$$

$$\cdot \text{ Original} = \frac{23177 \times 15}{2107} \times \frac{16}{15} \times \frac{15}{28}$$

$$3) -26.67\% = 1 - \frac{4}{15} \Rightarrow \frac{11}{15} \checkmark$$

$$\cdot \text{ Original} \Rightarrow \underline{18060}$$

	Part (Number)	Whole (Base)	Answer (%)
1.	204	425	48.0%
2.	147	735	20.0%
3.	258	215	120%
4.	224	672	33.33%
5.	104	640	16.25%
6.	103	927	11.11.0%
7.	196	784	25%
8.	116	638	18.18%
9.	97	679	14.29%
10.	396	924	42.86.0%
11.	64	832	7.69%
12.	603	268	225%
13.	483	420	115%
14.	546	468	116.66.0%
15.	931	399	233.33%

$$1) \frac{204}{425} \times 100$$

$$= 48.0\%$$

$$2) \frac{147}{735} \times 100$$

$$= 20.0\%$$

$$3) \frac{258}{x} \times 100 = 120$$

044

$$x \times 120.0\% = 258$$

$$x \times \frac{6}{5} = 258$$

$$x = 258 \times \frac{5}{6}$$

$$x = 215$$

$$4) \frac{x}{672} = 33.33.0\%$$

$$\left\{ 33.33.0\% = \frac{1}{3} \right\}$$

$$x = 672 \times \frac{1}{3}$$

$$x = 224$$

$$5) x \times 16.25\% = 104$$

$$x \times \frac{13}{80} = 104$$

$$x = \frac{104}{8} \times \frac{80}{13}$$

$$x = 640$$

$$11) x \times 7.69\% = 64$$

$$x \times \frac{1}{13} = 64$$

$$x = 64 \times 13$$

$$x = 832$$

$$6) \frac{1103}{927} \times 100 = 11.11\%$$

$$12) 268 \times 225\%$$

$$268 \times \frac{9}{4} = 603$$

$$7) x \times 25\% = 196$$

$$x \times \frac{1}{4} = 196$$

$$x = 196 \times 4$$

$$x = 784$$

$$13) x \times 115\% = 483$$

$$x \times \frac{23}{20} = 483$$

$$x = 483 \times \frac{20}{23}$$

$$x = 420$$

$$8) 638 \times 18.18\%$$

$$638 \times \frac{2}{11} = 116$$

$$14) \frac{546}{468} \times 100$$

$$= 116.66\%$$

$$9) 679 \times 14.29\%$$

$$679 \times \frac{1}{7} = 97$$

$$15) x \times 233.33\% = 931$$

$$x \times \frac{7}{3} = 931$$

$$x = 931 \times \frac{3}{7}$$

$$x = 399$$

$$10) \frac{396}{924} \times 100 = 42.86\%$$

(a)

(b)

	Number	Increase by	Increased Answer	Decreased by	Decreased Answer
1.	60	10%	66	20%	48
2.	80	25%		75%	
3.	400	38%	552	14%	344
4.	48	33.33%		62.50%	
5.	84	14.28%	96	16.67%	70
6.	52	7.69%		25%	
7.	240	125%	540	83.33%	40
8.	90	16.67%		91.67%	
9.	88	36.36%	120	75%	22
10.	108	41.67%			84
11.	120	62.5%	195	91.67%	10
12.	160		420		140
13.	75	12%	84	26.67%	55
14.	1280		2240		128
15.	630	77.77%	1120	42.86%	360
16.	960		1080		880
17.	136	5.88%	144	37.5%	85
18.	90		96	5.56%	
19.	225	26.67%	285	55.55%	100
20.	176	37.50%		90.91%	
21.	112	28.57%	144	92.86%	8

1) (a)

$$+ 10\% \Rightarrow 1 + \frac{1}{10} = \frac{11}{10}$$

$$60 \times \frac{11}{10} \Rightarrow 66 \checkmark$$

(b)

$$- 20\% \Rightarrow 1 - \frac{1}{5} = \frac{4}{5}$$

$$60 \times \frac{4}{5} \Rightarrow 48 \checkmark$$

2) (a)

$$+ 25\% \Rightarrow 1 + \frac{1}{4} = \frac{5}{4}$$

$$80 \times \frac{5}{4} \Rightarrow 100 \checkmark$$

(b)

$$- 75\% \Rightarrow 1 - \frac{3}{4} = \frac{1}{4}$$

$$80 \times \frac{1}{4} \Rightarrow 20 \checkmark$$

3) (a) $100\% + 38\% \Rightarrow 138\%$

$$400 \times \frac{138}{100} \Rightarrow 552 \checkmark$$

(b) $100\% - 14\% \Rightarrow 86\%$

$$400 \times 86\% \Rightarrow 344 \checkmark$$

4) (a) $+ 33.33\% \Rightarrow 1 + \frac{1}{3} = \frac{4}{3}$

$$48 \times \frac{4}{3} \Rightarrow 64 \checkmark$$

(b) $- 62.5\% \Rightarrow 1 - \frac{5}{8} = \frac{3}{8}$

$$48 \times \frac{3}{8} \Rightarrow 18 \checkmark$$

5) (a)

$$+14.28\% \Rightarrow 1 + \frac{1}{7} = \frac{8}{7}$$

$$84 \times \frac{8}{7} = \underline{\underline{96}} \checkmark$$

(b) $-16.67\% \Rightarrow 1 - \frac{1}{6} = \frac{5}{6}$

$$84 \times \frac{5}{6} = \underline{\underline{70}} \checkmark$$

8) (a)

$$+16.67\% \Rightarrow 1 + \frac{1}{6} = \frac{7}{6}$$

$$90 \times \frac{7}{6} \Rightarrow \underline{\underline{105}} \checkmark$$

(b) $-91.67\% \Rightarrow 1 - \frac{11}{12} = \frac{1}{12}$

$$90 \times \frac{1}{12} \Rightarrow \underline{\underline{7.5}} \checkmark$$

6) (a)

$$+7.69\% \Rightarrow 1 + \frac{1}{13} = \frac{14}{13}$$

$$\frac{52}{4} \times \frac{14}{13} \Rightarrow \underline{\underline{56}} \checkmark$$

(b) $-25\% \Rightarrow 1 - \frac{1}{4} = \frac{3}{4}$

$$\frac{52}{1} \times \frac{3}{4} \Rightarrow \underline{\underline{39}} \checkmark$$

9) (a) $88 \xrightarrow{+32} 120$

$$\Rightarrow \frac{32}{88} \times 100 = \underline{\underline{36.36\%}} \checkmark$$

(b) $88 \xrightarrow{-66} 22$

$$\Rightarrow \frac{66}{88} \times 100 = \underline{\underline{75\%}} \checkmark$$

7) (a)

$$+125\% \Rightarrow 1 + \frac{5}{4} = \frac{9}{4}$$

$$240 \times \frac{9}{4} \Rightarrow \underline{\underline{540}} \checkmark$$

(b) $-83.33\% \Rightarrow 1 - \frac{5}{6} = \frac{1}{6}$

$$240 \times \frac{1}{6} \Rightarrow \underline{\underline{40}} \checkmark$$

10) (a)

$$+41.67\% \Rightarrow 1 + \frac{5}{12} = \frac{17}{12}$$

$$108 \times \frac{17}{12} \Rightarrow \underline{\underline{153}} \checkmark$$

(b)

$$108 \xrightarrow{-24} 84$$

$$\frac{24}{108} \times 100 \Rightarrow \underline{\underline{22.22\%}} \checkmark$$

11) (a)

$$120 \xrightarrow{+75} 195$$

$$\frac{75}{120} \times 100 \Rightarrow \underline{62.5\%} \checkmark$$

(b)

$$-91.67\% \Rightarrow 1 - \frac{11}{12} = \frac{1}{12}$$

$$120 \times \frac{1}{12} = \underline{10} \checkmark$$

14) (a)

$$1280 \xrightarrow{+960} 2240$$

$$\frac{960}{1280} \times 100 \Rightarrow \underline{75\%} \checkmark$$

(b)

$$1280 \xrightarrow{-1152} 128$$

$$\frac{1152}{1280} \times 100 \Rightarrow \underline{90\%} \checkmark$$

12) (a)

$$160 \xrightarrow{+260} 420$$

$$\frac{260}{160} \times 100 \Rightarrow \underline{162.5\%} \checkmark$$

(b)

$$160 \xrightarrow{-20} 140$$

$$\frac{20}{160} \times 100 \Rightarrow \underline{12.5\%} \checkmark$$

15) (a)

$$630 \xrightarrow{+490} 1120$$

$$\frac{490}{630} \times 100 \Rightarrow \underline{77.77\%} \checkmark$$

(b)

$$-42.86\% \Rightarrow 1 - \frac{3}{7} = \frac{4}{7}$$

$$630 \times \frac{4}{7} \Rightarrow \underline{360} \checkmark$$

13) (a)

$$75 \xrightarrow{+9} 84$$

$$\frac{9}{75} \times 100 \Rightarrow \underline{12\%} \checkmark$$

(b)

$$-26.67\% = 1 - \frac{4}{15} \Rightarrow \frac{11}{15}$$

$$75 \times \frac{11}{15} \Rightarrow \underline{55} \checkmark$$

16) (a)

$$960 \xrightarrow{+120} 1080$$

$$\frac{120}{960} \times 100 \Rightarrow \underline{12.5\%} \checkmark$$

(b)

$$960 \xrightarrow{-80} 880$$

$$\frac{80}{960} \times 100 \Rightarrow \underline{8.33\%} \checkmark$$

17) (a)

$$136 \xrightarrow{+8} 144$$

$$\frac{18}{136} \times 100 \Rightarrow \underline{5.88\%}$$

(b)

$$136 \xrightarrow{-51} 85$$

$$\frac{51}{136} \times 100 \Rightarrow \underline{37.5\%}$$

18) (a)

$$90 \xrightarrow{+6} 96$$

$$\frac{6}{90} \times 100 \Rightarrow \underline{6.66\%}$$

(b) $-5.56\% \Rightarrow 1 - \frac{1}{18} = \frac{17}{18}$

$$5 \times 90 \times \frac{17}{18} \Rightarrow \underline{85} \checkmark$$

19) (a)

$$225 \xrightarrow{+60} 285$$

$$\frac{60}{225} \times 100 \Rightarrow \underline{26.67\%}$$

(b) $-55.55\% \Rightarrow 1 - \frac{5}{9} = \frac{4}{9}$

$$225 \times \frac{4}{9} \Rightarrow \underline{100} \checkmark$$

20) (a)

$$-37.5\% \Rightarrow 1 - \frac{3}{8} \Rightarrow \frac{5}{8}$$

$$176 \times \frac{5}{8} \Rightarrow \underline{110} \checkmark$$

(b) $-90.90\% \Rightarrow 1 - \frac{10}{11} = \frac{1}{11}$

$$176 \times \frac{1}{11} \Rightarrow \underline{16} \checkmark$$

21) (a)

$$+28.57\% \Rightarrow 1 + \frac{2}{7} \Rightarrow \frac{9}{7}$$

$$112 \times \frac{9}{7} \Rightarrow \underline{144} \checkmark$$

(b)

$$-92.86\% \Rightarrow 1 - \frac{13}{14} \Rightarrow \frac{1}{14}$$

$$112 \times \frac{1}{14} \Rightarrow \underline{8} \checkmark$$